

3.3 Worksheet

Determine whether the following series converge or diverge. Use any test or idea that is helpful from any section from 3.1 – 3.3.

1. $\sum_{k=1}^{\infty} \frac{7}{3k}$

We note that: $\sum_{k=1}^{\infty} \frac{7}{3k} = \frac{7}{3} \cdot \sum_{k=1}^{\infty} \frac{1}{k}$. Since $\sum_{k=1}^{\infty} \frac{1}{k}$ is the harmonic series it diverges and therefore the given series diverges. ■

2. $\sum_{k=1}^{\infty} \frac{4k^3 + 2k}{2k^3 - 1}$

Let $a_k = \frac{4k^3 + 2k}{2k^3 - 1}$ and then $f(x) = \frac{4x^3 + 2x}{2x^3 - 1}$. We note that

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{4x^3 + 2x}{2x^3 - 1} = \lim_{x \rightarrow \infty} \frac{4x^3 + 2x}{2x^3 - 1} \cdot \frac{\left(\frac{1}{x^3}\right)}{\left(\frac{1}{x^3}\right)} = \lim_{x \rightarrow \infty} \frac{4 + \frac{2}{x^2}}{2 - \frac{1}{x^3}} = 2 \text{ and thus } \lim_{k \rightarrow \infty} a_k = 2. \text{ Therefore (since}$$

the limit is not equal to 0) the series diverges by the Test for Divergence. ■

3. $\sum_{n=1}^{\infty} \arctan(n)$

We know that $\lim_{n \rightarrow \infty} \arctan(n) = \frac{\pi}{2}$. Therefore, (since the limit is not equal to 0), the series diverges by the Test for Divergence. ■

4. $\sum_{n=1}^{\infty} \frac{\pi^k}{3^{2k+1}}$

$\sum_{n=1}^{\infty} \frac{\pi^k}{3^{2k+1}} = \sum_{n=1}^{\infty} \frac{\pi^k}{3 \cdot 9^k} = \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{\pi}{9}\right)^k = \sum_{n=1}^{\infty} \frac{\pi}{27} \left(\frac{\pi}{9}\right)^{k-1}$. This is a geometric series with $r = \frac{\pi}{9}$ and since $|r| < 1$

this series converges. $\sum_{n=1}^{\infty} \frac{\pi}{27} \left(\frac{\pi}{9}\right)^{k-1} = \frac{\frac{\pi}{27}}{1 - \left(\frac{\pi}{9}\right)} = \frac{\frac{\pi}{27}}{\frac{9 - \pi}{9}} = \frac{9\pi}{27(9 - \pi)} = \frac{\pi}{3(9 - \pi)}$. ■

5. $\sum_{n=1}^{\infty} \sqrt[n]{2}$

We know that $\lim_{n \rightarrow \infty} \sqrt[n]{2} = \lim_{n \rightarrow \infty} 2^{1/n} = 2^0 = 1$. Therefore, (since the limit is not equal to 0), the series diverges by the Test for Divergence. ■

6. $\sum_{n=1}^{\infty} \frac{1+2^n}{3^n}$

We first see that $\sum_{n=1}^{\infty} \frac{1+2^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{1}{3^n} + \frac{2^n}{3^n} \right)$. We want to split this series into two series, but we need to first investigate each series separately to ensure that they converge.

Note that $\sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3} \right)^{n-1}$ which is a geometric series with $r = \frac{1}{3}$ and since $|r| < 1$ this series converges.

Note that $\sum_{n=1}^{\infty} \frac{2^n}{3^n} = \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{2}{3} \right)^{n-1}$ which is a geometric series with $r = \frac{2}{3}$ and since $|r| < 1$ this series converges.

Therefore, since both series converge we can say that

$$\sum_{n=1}^{\infty} \left(\frac{1}{3^n} + \frac{2^n}{3^n} \right) = \sum_{n=1}^{\infty} \frac{1}{3^n} + \sum_{n=1}^{\infty} \frac{2^n}{3^n} = \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3} \right)^{n-1} + \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{2}{3} \right)^{n-1} = \frac{1/3}{1-(1/3)} + \frac{2/3}{1-(2/3)} = \frac{1/3}{2/3} + \frac{2/3}{1/3} = \frac{1}{2} + 2 = 2.5$$

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7. $\sum_{n=1}^{\infty} \frac{n-1}{3n-1}$

Let $f(x) = \frac{x-1}{3x-1}$. We note that $\lim_{x \rightarrow \infty} f(x) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{3} = \frac{1}{3}$ and therefore $\lim_{n \rightarrow \infty} \frac{n-1}{3n-1} = \frac{1}{3}$.

Therefore, (since the limit is not equal to 0), the series diverges by the Test for Divergence. ■

8. $\sum_{n=4}^{\infty} \frac{1}{n-1}$

Note that $\sum_{n=4}^{\infty} \frac{1}{n-1} = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$. This series is the harmonic series with two terms removed. Since only a finite number of terms was removed this series is still divergent. ■

9. $\sum_{n=1}^{\infty} \left(\frac{1}{e^n} + \frac{1}{n} - \frac{1}{n+1} \right)$

We want to split this series into two series, but we need to first investigate each series separately to ensure that they converge.

Note that $\sum_{n=1}^{\infty} \frac{1}{e^n} = \sum_{n=1}^{\infty} \frac{1}{e} \left(\frac{1}{e} \right)^{n-1}$ which is a geometric series with $r = \frac{1}{e}$ and since $|r| < 1$ this series converges.

Note that $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$ appears to be a telescoping series, but we need to find s_k .

Note that $s_k = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{k+1}$. We note that $\lim_{k \rightarrow \infty} \left(1 - \frac{1}{k+1} \right) = 1$ and therefore $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$ converges to 1.

Therefore, since both series converge we can say that

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{1}{e^n} + \frac{1}{n} - \frac{1}{n+1} \right) &= \sum_{n=1}^{\infty} \frac{1}{e^n} + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \frac{1}{e} \left(\frac{1}{e} \right)^{n-1} + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{1/e}{1-(1/e)} + 1 \\ &= \frac{1/e}{\left(\frac{e-1}{e} \right)} + 1 = \frac{1}{e-1} + 1 = \frac{e}{e-1} \end{aligned}$$

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10. If $\sum a_n$ is convergent and $\sum b_n$ is divergent, show that the series $\sum (a_n + b_n)$ is divergent. *Hint:* Argue by first *assuming* that $\sum (a_n + b_n)$ is convergent and using the series $\sum (-a_n)$ to help show that this assumption is impossible.

Let's assume that $\sum (a_n + b_n)$ is convergent. Since $\sum a_n$ is convergent then $\sum (-a_n)$ must also be convergent. However, this also means that the series $\sum [(a_n + b_n) + (-a_n)]$ is also convergent (since it is the sum of two convergent series). But, we see that $\sum [(a_n + b_n) + (-a_n)] = \sum b_n$ must then be convergent, but this is a contradiction since we are told that $\sum b_n$ is divergent. Therefore, we know that $\sum (a_n + b_n)$ must be divergent.

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11. $\sum_{n=1}^{\infty} \left(\frac{2}{n+1} - \frac{3}{5^n} \right)$ *Hint: Consider the work from question #10.*

We want to split this series into two series, but we need to first investigate each series separately to ensure that they converge.

Note that $\sum_{n=1}^{\infty} \frac{2}{n+1} = 2 \sum_{n=1}^{\infty} \frac{1}{n+1}$ and $\sum_{n=1}^{\infty} \frac{1}{n+1}$ is just the Harmonic series with a single term missing.

Therefore, this series diverges.

Note that $\sum_{n=1}^{\infty} \frac{3}{5^n} = \sum_{n=1}^{\infty} \frac{3}{5} \left(\frac{1}{5} \right)^{n-1}$ which is a geometric series with $r = \frac{1}{5}$ and since $|r| < 1$ this series converges.

Therefore, based on what we showed in question #10 the series $\sum_{n=1}^{\infty} \left(\frac{2}{n+1} - \frac{3}{5^n} \right)$ must diverge.

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